

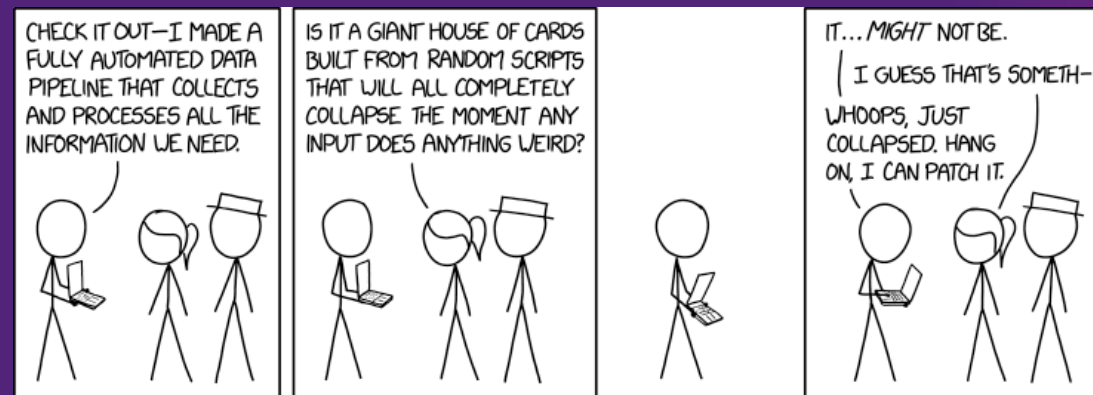


REIT6811 | Research Methods

Lecture 6: Data, Analysis & Data Visualization

Professor Markus Barth

Slides adapted from:
Dr Monique Tourell



Acknowledgement of Country

I acknowledge the Turrbal and Yuggera peoples as the Traditional Owners and custodians of Meanjin, the lands on which this course was developed and where we meet today.

I pay my respects to their Ancestors and their descendants, who continue cultural and spiritual connections to Country.

I recognize that these lands have been a place of teaching, learning, research and collaboration for tens of thousands of years before the establishment of a UQ campus.

Sovereignty was never ceded.

The Brisbane River pattern from A Guidance Through Time by Casey Coolwell and Kyra Mancktelow.



Research

- *Research is a decision-making journey*
- You need to constantly **engage in decision-making** that is logical, consistent, and coherent
- The choices you make must take you one step **closer to being able to answer your research question** credibly
- Without clear articulation of your research question, you are travelling blind

Source: O'Leary, Z (2018). Little quick fix: Research Question. SAGE, Los Angeles.

Research Questions

Once you've completed your Literature Review and found a knowledge gap **you need to develop a research question:**

Knowledge Gap	Research Question
Current deep learning models require vast amounts of labeled data to perform well, but there is limited understanding of how to efficiently train models with minimal labeled data.	How can semi-supervised learning techniques be improved to reduce the dependency on labeled data in deep learning models while maintaining or improving performance?
Despite advancements in encryption, there is a lack of effective techniques to ensure secure communication in quantum computing environments.	What are the most effective cryptographic methods for ensuring secure communication in a quantum computing environment, and how can they be implemented practically?
The generalization of computer vision models across different environments and lighting conditions remains a challenge.	What computer vision models currently perform the best in varying environmental and lighting conditions, and what techniques can be developed to improve their robustness?

Developing Research Questions

- A common criteria for what defines a research question is that it must be **'researchable'** (White 2009, p. 35).
- Silverman (2001) says research questions should be able to say **'a lot about a little'** rather than **'a little about a lot'**.
- 'Research questions should also be able to generate **knowledge that matters to society**' (Alvesson and Sandberg 2013: p.12).
- 'You should be sure that your questions are formulated in such a way that **intellectually interesting answers and arguments are possible and probable**' (Mason, 2018, p. 15)

White, P. (2009) Developing Research Questions: A Guide for Social Scientists. Michigan: Palgrave.

Silverman, D. (2001) Interpreting Qualitative Data 2 Ed. London: Sage.

Alvesson M. and Sköldberg K. (2000) Reflexive Methodology: New Vistas for Qualitative Research. London: Sage.

Mason, J. 2018 Qualitative Researching, University of Manchester, UK: Sage

Developing Research Questions

The PICO method is a framework for developing research questions that is commonly used in clinical/medical research.

- ✓ **P**opulation: What is research population (research object)?
- ✓ **I**ntervention: What is the main treatment or diagnostic test (theory/method)?
- ✓ **C**omparison: Alternative strategy/treatment to compare to (theories/methods)?
- ✓ **O**utcome: Result of intervention/knowledge generation

Developing Research Questions

The PICO method is a framework for developing research questions which can be adapted for other disciplines:

Population

Deep-learning models

Intervention

Semi-supervised learning techniques

Comparison

Fully-supervised learning techniques

Outcome

Reduced dependency on labelled data, improved performance

Research Question

How can semi-supervised learning techniques be improved to reduce the dependency on labeled data in deep learning models while maintaining or improving performance?

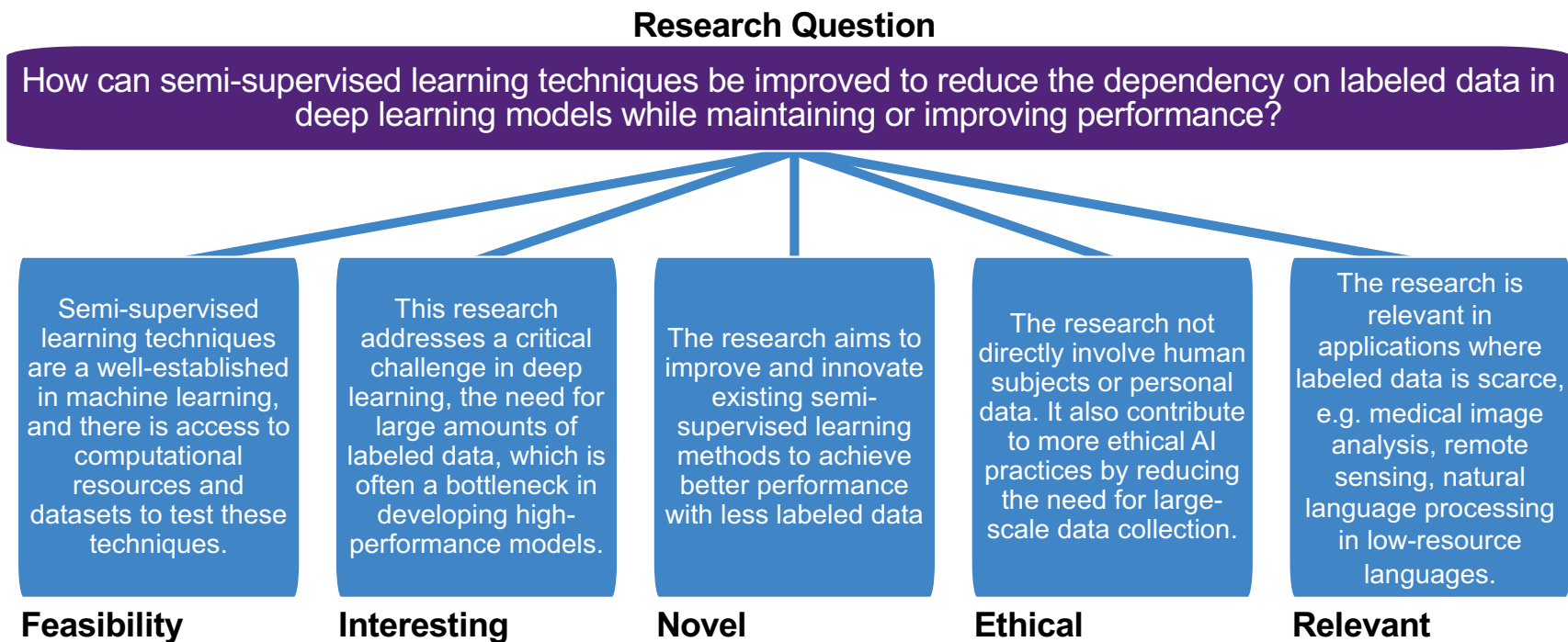
Developing Research Questions

The FINER method is a framework for evaluating and refining research questions:

- ✓ **Feasible:** Can I realistically conduct this research with the resources and time available?
- ✓ **Interesting:** Am I and others genuinely interested in the outcome of this research?
- ✓ **Novel:** Does this research offer something new to the field?
- ✓ **Ethical:** Does this research comply with ethical standards?
- ✓ **Relevant:** Will the results of this research be valuable to others in the field or society at large?

Developing Research Questions

The FINER method is a framework for evaluating and refining research questions:



Research in Engineering/Computer Science

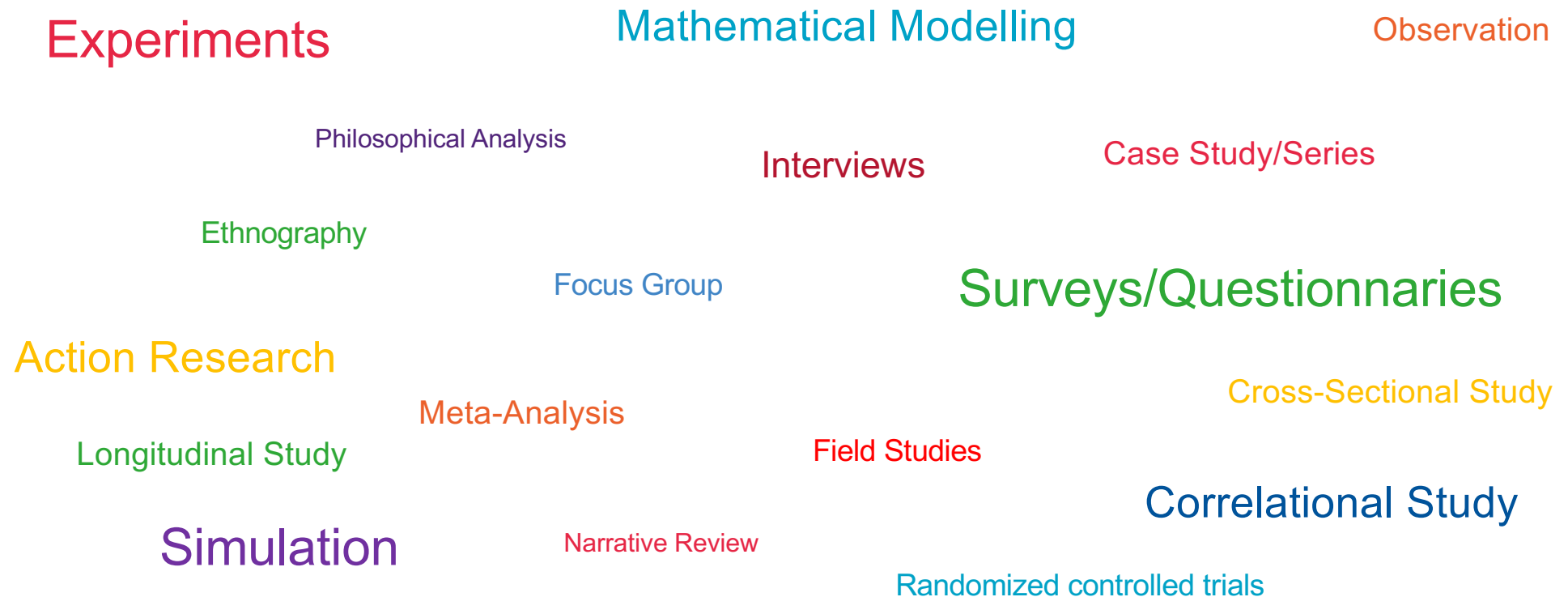
- Developing new algorithm/system/device
- Anyone can create some new thing, but one must show that it is “better”
- Research Questions might try to:
 - Solves a problem in less time or more accurately
 - Solves a larger class of problems
 - Is more efficient in resources
 - Is more expressive by some criterion
 - Is more visually appealing



Journal Activity

Are all the criteria in the PICO and FINER frameworks relevant all of the time? Can you think of some scenarios where these frameworks wouldn't work?

Research Methodologies



Research Methodologies

Action research

- Pursues action and understanding at the same time
- Continuously alternates between action and critical reflection, while refining methods, data and interpretation in the light of the understanding developed in the earlier cycles

Example: Iterative software development and testing

Case study

- In-depth exploration of a specific real-life situation or individual
- Usually generates a large amount of (subjective) data
- Should not merely report the data obtained or behaviour observed but attempt to generalise from the specific details of the situation observed

Example: Case study of open-source software development

Source: Ullrich Hustadt "Research Methods in Computer Science" <https://cgi.csc.liv.ac.uk/~ullrich/COMP516/notes/lect06.pdf>

Research Methodologies

Survey

- Usually via questionnaires or interviews
- Questionnaire and interview design are important!
- Determination of sample size and sample elements important!

Example: Survey on the popularity of programming languages

Experiment

- Investigation of causal relationships using test controlled by the researcher
- Usually performed in development, evaluation and problem-solving projects

Example: Evaluation of processor performance

Source: Ullrich Hustadt "Research Methods in Computer Science" <https://cgi.csc.liv.ac.uk/~ullrich/COMP516/notes/lect06.pdf>

Purpose of Data in Research

Your choice of research methodology will depend on the research question, but will almost always involve the collection and analysis of data to **draw conclusions from empirical evidence**

- Data enables researchers to:
 - Answer questions
 - Test hypotheses
 - Validate theories
 - Gain insights into phenomena, patterns, trends, and relationships
- Data and analysis contributes new knowledge or understanding to a particular field, inform decision-making processes, and advance scientific discovery

Types of Data and Analysis

Usually distinguish between three different types of data in research:

- Quantitative
- Qualitative
- Mixed methods

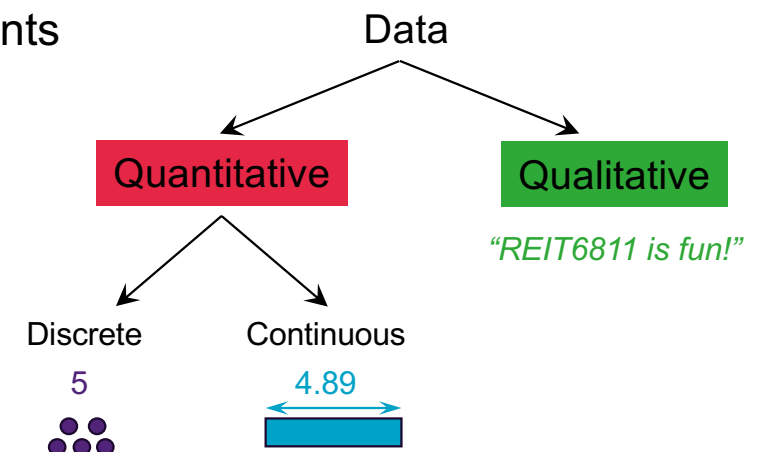
Quantitative and Qualitative Research Methods

Quantitative:

- Collects **numerical data** that can be measured and represented numerically (counts, ratings, measurements)
- Analyses patterns and relationships between variables through statistical analysis, trends, causal effects
- Objective or subjective, typically collected through experiments (sensors, scales) surveys, simulations

Qualitative:

- **Non-numerical data** (e.g., words, images, video)
- Explore subjective experiences, opinions and attitudes
- Often acquired through observation and interviews



Quantitative vs Qualitative Research

	Qualitative	Quantitative
Nature of data	Visual and verbal recording in rich detail	Responses distilled into numeric scores
Relevance of context	Results are generally assumed to be specific to time, place, people, and culture studied	Results are generally assumed to be generalisable across contexts and cultures
Nature and control of potential causes	Ideally naturalistic with multiple factors shaping the behaviours observed and discussed	Ideally settings are controlled and variables are manipulated or measured to allow causal inference
Key research instrument	The researcher is the instrument and uses skills and rapport to gain insights based on trust	Researcher tries to be invisible and relies on responses to structured measures or choices

Source: Belk, R., Fischer, E. & Kozinets, R.V. (2013). *Qualitative Consumer & Marketing Research*, Sage: Los Angeles, p. 3

Quantitative and Qualitative Examples

Research Question: How does lecture attendance rate affect student grades?

Objective Quantitative – correlational study

Collect data on student attendance rate and final grades for different courses

Student Lecture Attendance Rate (%)	Student Final Grade (%)
90	88
75	72
50	62
80	81
100	85
70	54
30	65

Subjective Quantitative - closed survey

My attendance at lectures has influenced my grades at university.

- 4 — Strongly Agree
- 3 — Agree
- 2 — Disagree
- 1 — Strongly Disagree

Qualitative – open survey/interview

In what ways do you think your attendance at lectures has influenced your grades at university?

Quantitative Research Methods

Purpose:

- Measure differences, assess (causal) relationships between variables, test hypotheses, find patterns, or create a set of observations
- Generalise results to wider populations or broader contexts
- Usually inductive reasoning – make generalized conclusions based off specific observations

Data collection and reporting findings:

- Data collection through experiments, surveys, measurements
- Accuracy and repeatability are important

Quantitative Data Analysis

Some factors affecting how you analyse data:

- Research question – what do you want to know?
- Data type (continuous, discrete)
- Data shape (normal, skewed)
- Sample size

Some Data Analysis Methods:

- Descriptive statistics (mean, median, etc.)
- Inferential statistics (ANOVA, t-tests, regression analysis)
- Fits to mathematical models
- Quality Metrics (structural similarity index, mortality rates)
- Error Analysis

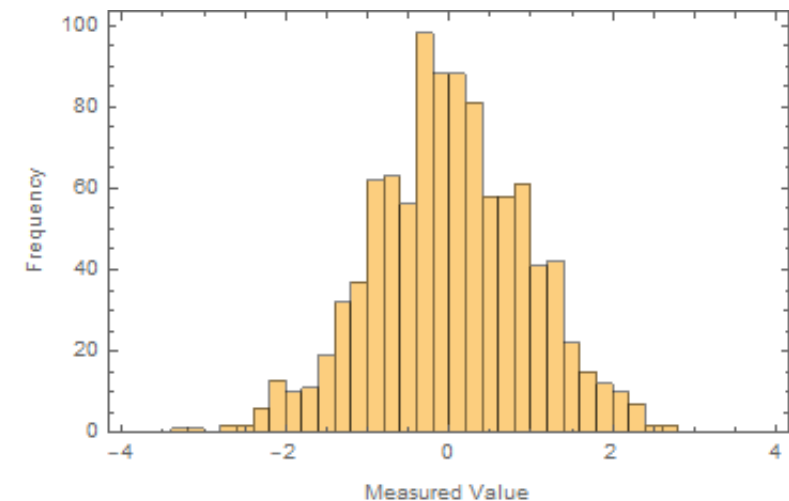
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Describe and summarize aspects of the data

Mean = 0.01

Median = -0.04

Standard Deviation = 0.96

Min = -3.26

Max = 2.80

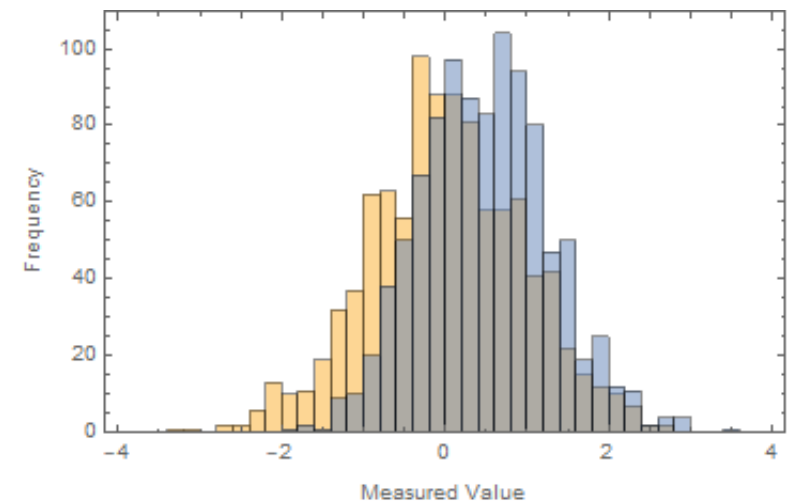
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Make inferences and generalizations

Hypothesis testing – comparing data to a control

Regression Analysis – assess the correlation between variables

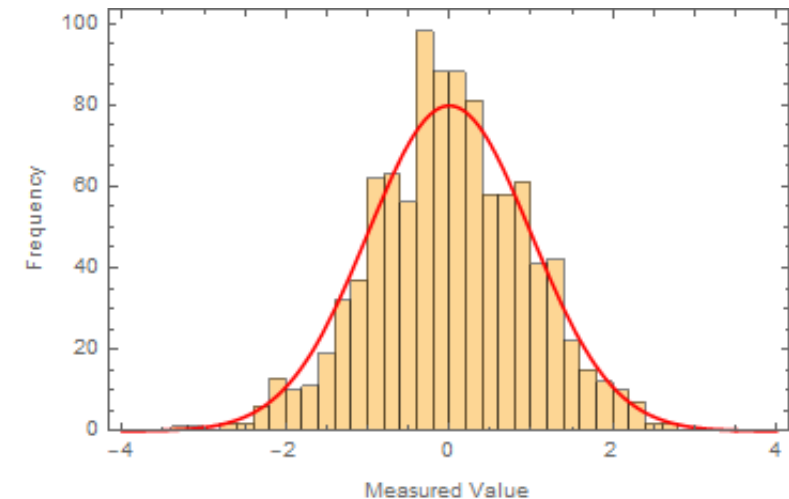
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Test theories and models

Compare data to principles-derived theories

Validate or refine mathematical models

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Feng, Ruimin, et al. "MoDL-QSM: Model-based deep learning for quantitative susceptibility mapping." *NeuroImage* 240 (2021): 118376.

Table 1. Quantitative performance metrics, RMSE, SSIM, and HFEN from the five different QSM reconstruction methods referenced to the label χ_{33} . MoDL-QSM shows better performances in all criteria than other methods.

	RMSE	SSIM	HFEN
TKD	92.42±3.99	0.7753±0.0165	90.23±3.65
STAR-QSM	79.57±3.27	0.8180±0.0160	77.26±2.81
AutoQSM	69.40±2.68	0.8549±0.0193	70.04±2.73
QSMnet	55.07±3.24	0.8759±0.0141	54.39±2.70
MoDL-QSM	54.86±3.10	0.8801±0.0140	54.31±2.87

Derive standard-based quality indicators

Compare to other data/methods

Measure performance against benchmarks

Refine metrics for improvement

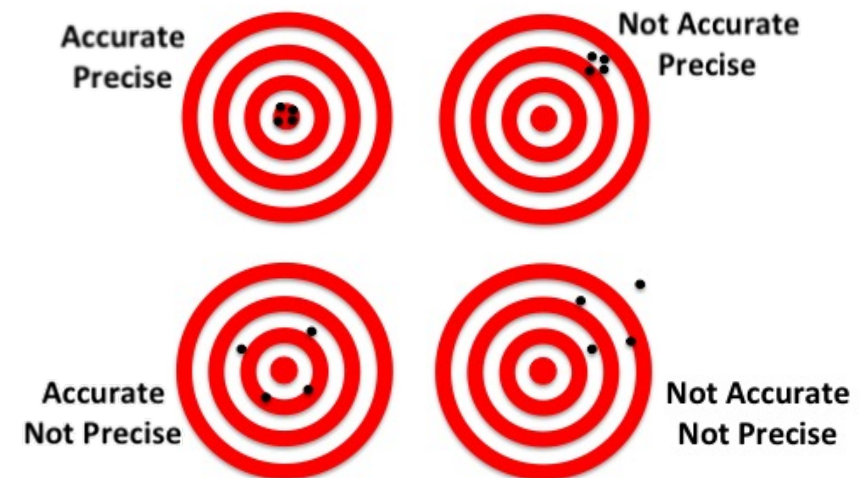
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Identify and categorize errors

Determine the source/cause of errors

Determine the impact of errors on conclusions

Refine methods to reduce errors

Quantitative Data Analysis

Visualization of Data:

- Tables
- Graphs and Charts (scatter, histograms, box plot, bubble, ridgeline, pie chart, correlogram)
- Maps (geographic, value)
- Spectra

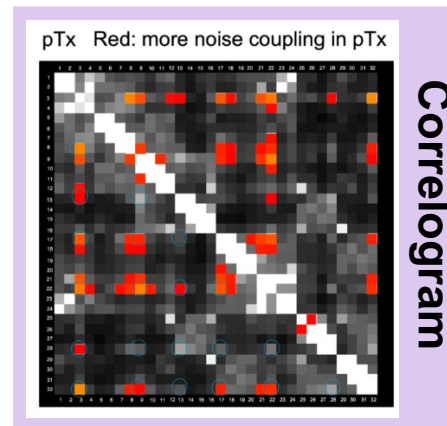
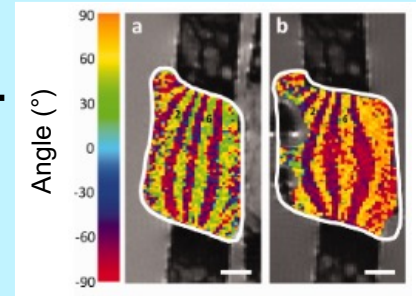
Software:

R, Python, SBSS, Excel, Matlab, Mathematica

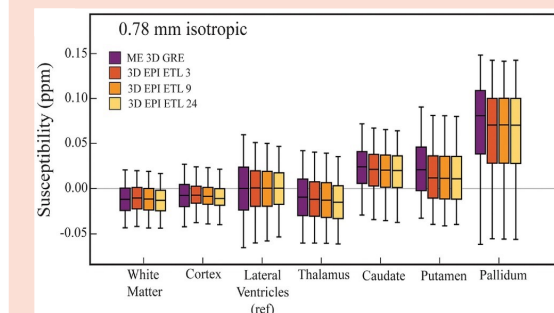
Check out the UQ Library Guide:

<https://guides.library.uq.edu.au/research-techniques/data-visualisation>

Value Maps

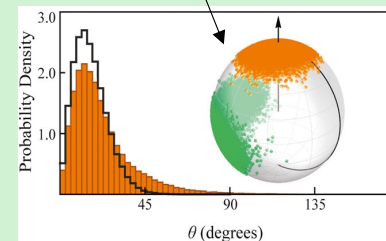


Correlogram

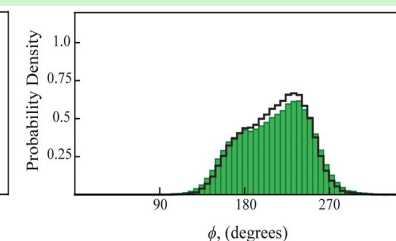


Box Plot

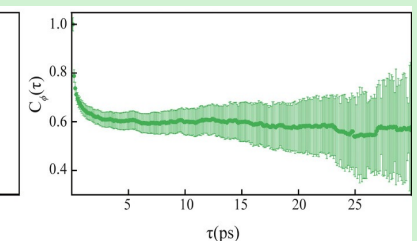
Scatter on sphere



Histograms



Scatter with error





Journal Activity

Datawrapper activity to prep for Tutorial 5: chance to practice working with and analysing both quantitative and qualitative data

Under Lecture Slides on Blackboard, you will find a .csv file of data with student attendance at lectures and student final grade.

Use whatever software you are familiar with (Excel, MATLAB, Python, etc.) to create an appropriate plot of the data

<https://app.datawrapper.de>

Qualitative Research Methods

Purpose:

- Exploring and understanding subjective experiences, beliefs, and meanings

Data collection and reporting findings:

- Interviews, focus groups, case studies, grounded theory and observations
- Findings presented in narrative form, often with quotes and examples

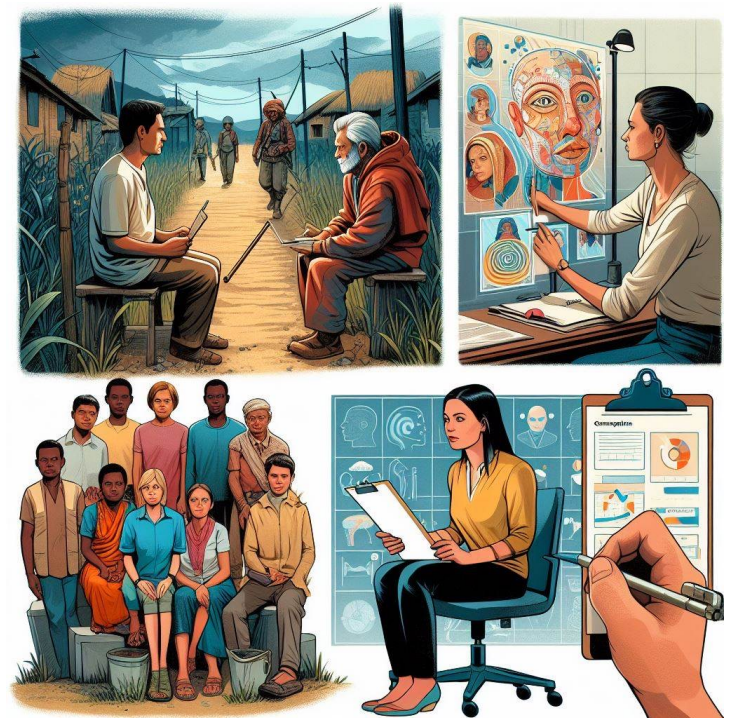


Image created by OpenAI's DALL-E3 via Bing, accessed 5/04/24.

Qualitative Research Methods

Examples:

- Ethnographic studies of social interactions
- Case studies of individual experiences
- Qualitative interviews with experts on a particular topic
- Analysis of textual data from documents or interviews

Factors affecting how you analyse data:

- Research question – what do you want to know?
- Data type (structured, unstructured, etc.)
- Sample size

Qualitative Data Analysis

Data analysis methods:

- Pattern finding
- Coding and thematic analysis
- Deriving quantitative values

Visualising:

- Word clouds, conceptual maps

Software:

- Nvivo & Leximancer
- <https://my.uq.edu.au/information-and-services/information-technology/software-and-web-apps/free-student-software-personal-use>

NVivo	Free for academic use.	Windows/Mac	Complete the Request a Licence form. Note this licence expires 19/11/2022.
Leximancer	Free for academic use.	Windows/Mac	Complete the Request a Licence form.

Qualitative Data Analysis - Thematic analysis

Examining commonality, examining differences and examining relationships in data (Gibson and Brown 2009, pp. 128-129)

Codes vs themes:

- A 'theme is an outcome of coding, categorization, or analytic reflection, not something that is, in itself, coded' (Saldana, 2013, p.14)
- Themes should emerge as you consider the **relationship between codes** (e.g. why some codes occur, and others don't) and **themes**



Qualitative Data Analysis – Example Thematic Analysis

Table: Example presentation of themes, codes and quotations in a qualitative findings section

Themes	Codes	Examples
Drinking to socialize	Social environment	"I drink about half the time I'm in a big social environment. Like a house party I'll usually drink, besides that it doesn't matter."
Drinking to relieve stress	Week activities	"It was such a stressful week because we had been planning and there was more alcohol than I thought there would be."
Drinking because of curiosity	Finding out for myself	"I am gonna find out for myself. So I did."
Drinking to feel a sense of community	Connections	"You kind of have like this weird connection with everyone else that you're with."
Drinking to relax	Before sleep	"I'll have a gin and tonic or something before I go to bed sometimes. It really helps. It's relaxing."

Source: Creswell, 2013, p. 180

Mixed Methods

Mixed methods research is employed when both qualitative and quantitative data will better inform the analysis in order to answer the research questions at hand.

Each **paradigm** contributes its own strengths but compensates for the other paradigm's information-generating deficiencies.

Tutorial 4: You will analyze a mixed method data set and hopefully see this first-hand.

Source: Saldaña & Omasta, Qualitative Research: Analyzing Life. SAGE Publishing, 2018

Criteria for evaluating research and evidence

Quantitative Lens	Qualitative Lens	Bridge	Essential Questions	Morse (Chapter 2)
<i>Validity</i>	Credibility	Truth	Are the results believable, seem truthful?	Encapsulates the idea that you need to be able to demonstrate that your concepts can be identified, observed or 'measured' in the way you say they can. <i>Work out how well a particular method and data source might illuminate your concepts</i>
<i>Reliability</i>	Dependability	Replicable	Are results consistent over time?	Involves the accuracy of your research methods and techniques. <i>How reliably and accurately do they produce data? How can you maximize their reliability?</i>
<i>Generalizability</i>	Transferability	Applicable	Are the results applicable to similar settings?	Generalizability involves the extent to which you can make some form of wider claim on the basis of your research and analysis, rather than simply stating that your analysis is entirely idiosyncratic and particular. <i>You need to ensure that you are thinking about the basis on which you can make general claims, as well as what kinds of general claims your research questions, epistemological approach and intellectual puzzle might imply</i>
Objectivity	Confirmability	Neutrality	Are results corroborated via triangulation?	
Accuracy	Authenticity	Reality(ies)	Are all 'realities' represented?	

Source: Adapted from Previte, J. RBUS6904 University of Queensland, Morse (2018); Bilups (2020);

Data Analysis – Challenges

Data quality

- Data errors, inconsistencies, missing values, outliers

Biases

- Systematic errors or distortions in the research process that can influence the findings
- Can occur during participant selection, data collection, or data analysis

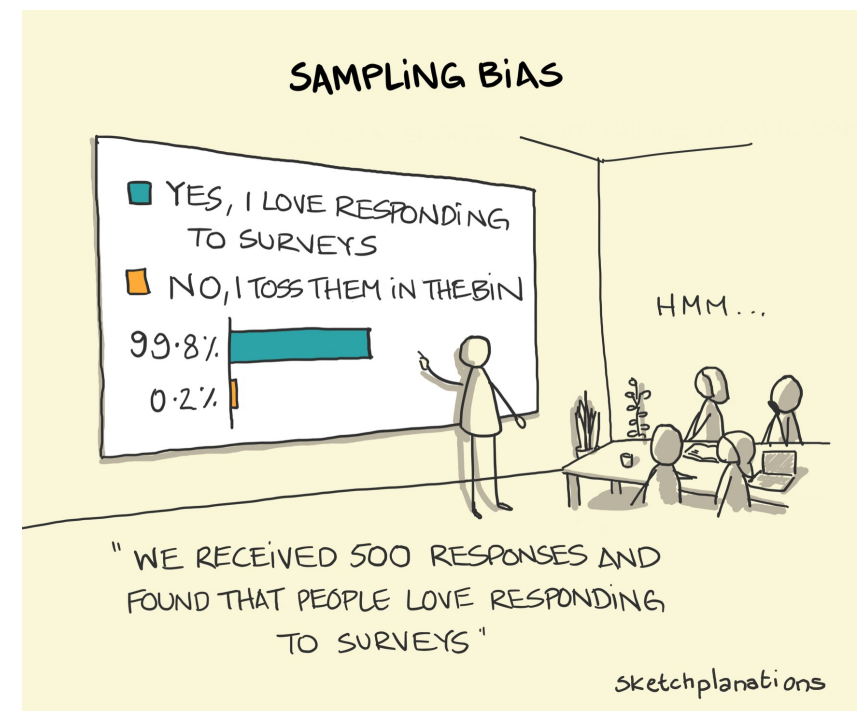


Image: <https://sketchplanations.com/sampling-bias>

Data Analysis – Challenges

Sample size

- Greater sample size = greater statistical power, making it more likely to detect a significant difference
- Can estimate required sample size based on the desired level of *confidence* and expected *effect size* using statistical power analysis tools

Statistical assumptions

- Failure to meet assumptions (normality, homogeneity of variance) required for valid statistical analysis, can lead to incorrect conclusions or biased results

Subjectivity

- Influence of the researcher's personal feelings, values, beliefs, and experiences on the research process and findings

Data Analysis – Challenges

Ethical concerns

- Adhere to established ethical guidelines set by professional organizations and regulatory bodies
- Unethical data analysis practices
 - Data falsification
 - P-hacking
 - Selective reporting



Thinking about evidence, and vice versa

<https://datacolada.org/>



Australian Computer Society Code of Professional Conduct

1. **The Primacy of the Public Interest** *You will place the interests of the public above those of personal, business or sectional interests.*

https://www.acs.org.au/content/dam/acs/Code-of-Professional-Conduct_v2.1.pdf

AI Ethics

Human, societal and environmental wellbeing

AI systems should benefit individuals, society and the environment.

Human-centred values

AI systems should respect human rights, diversity, and the autonomy of individuals.

Fairness

AI systems should be inclusive and accessible, and should not involve or result in unfair discrimination against individuals, communities or groups.

Privacy protection and security

AI systems should respect and uphold privacy rights and data protection, and ensure the security of data.



Reliability and safety

AI systems should reliably operate in accordance with their intended purpose.

Transparency and explainability

There should be transparency and responsible disclosure so people can understand when they are being significantly impacted by AI, and can find out when an AI system is engaging with them.

Contestability

When an AI system significantly impacts a person, community, group or environment, there should be a timely process to allow people to challenge the use or outcomes of the AI system.

Accountability

People responsible for the different phases of the AI system lifecycle should be identifiable and accountable for the outcomes of the AI systems, and human oversight of AI systems should be enabled.

<https://www.industry.gov.au/publications/australias-artificial-intelligence-ethics-framework/australias-ai-ethics-principles>

Summary

- **Research:**
 - Research questions and methodology
 - Experimental hypotheses
 - Evaluating research
- **Data:**
 - Used to draw conclusions from empirical evidence
 - Quantitative, qualitative and mixed methods
 - Challenges
 - Ethics
 - Presentation of data

Machine learning resources

Repeatability is important

Datasets and code repositories:

- <https://paperswithcode.com/> [machine learning papers, code, datasets, methods and evaluation tables]
- <https://huggingface.co/> [machine learning community collaborates on models, datasets, and applications]